E-Commerce Sales Analysis Project

```
import pandas as pd
import plotly.express as px
import plotly.graph_objects as go
import plotly.io as pio
import plotly.colors as colors
pio. templates.default ="plotly_white"

pip install --upgrade plotly

Defaulting to user installation because normal site-packages is not
writeable
Requirement already satisfied: plotly in c:\programdata\anaconda3\lib\
site-packages (5.24.1)
Collecting plotly
  Downloading plotly-6.5.0-py3-none-any.whl.metadata (8.5 kB)
Collecting narwhals>=1.15.1 (from plotly)
  Downloading narwhals-2.12.0-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: packaging in c:\programdata\anaconda3\
lib\site-packages (from plotly) (24.1)
Downloading plotly-6.5.0-py3-none-any.whl (9.9 MB)
   ---------------------------------------- 0.0/9.9 MB ? eta -:--:--
   -------- ------------------------------- 2.1/9.9 MB 14.7 MB/s eta
0:00:01
   ------------------ --------------------- 4.5/9.9 MB 12.2 MB/s eta
0:00:01
   ---------------------- ----------------- 5.5/9.9 MB 9.9 MB/s eta
0:00:01
   ------------------------- -------------- 6.3/9.9 MB 8.8 MB/s eta
0:00:01
   ----------------------------- ---------- 7.3/9.9 MB 7.7 MB/s eta
0:00:01
   ---------------------------------- ----- 8.7/9.9 MB 7.3 MB/s eta
0:00:01
   -------------------------------------- - 9.4/9.9 MB 6.5 MB/s eta
0:00:01
   ---------------------------------------  9.7/9.9 MB 6.2 MB/s eta
0:00:01
   ---------------------------------------- 9.9/9.9 MB 5.8 MB/s eta
0:00:00
Downloading narwhals-2.12.0-py3-none-any.whl (425 kB)
Installing collected packages: narwhals, plotly
Successfully installed narwhals-2.12.0 plotly-6.5.0
Note: you may need to restart the kernel to use updated packages.

  WARNING: The script plotly_get_chrome.exe is installed in 'C:\Users\
asusl\AppData\Roaming\Python\Python312\Scripts' which is not on PATH.
```

```
  Consider adding this directory to PATH or, if you prefer to suppress
this warning, use --no-warn-script-location.
```

```python
import pandas as pd

# Read CSV file
df = pd.read_csv("Sample - Superstore.csv", encoding='latin-1')

# Optional: show first few rows
df.head()
```

```
   Row ID        Order ID  Order Date   Ship Date       Ship Mode
Customer ID  \
0       1  CA-2016-152156   11/8/2016  11/11/2016    Second Class
CG-12520
1       2  CA-2016-152156   11/8/2016  11/11/2016    Second Class
CG-12520
2       3  CA-2016-138688   6/12/2016   6/16/2016    Second Class
DV-13045
3       4  US-2015-108966  10/11/2015  10/18/2015  Standard Class
SO-20335
4       5  US-2015-108966  10/11/2015  10/18/2015  Standard Class
SO-20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID         Category Sub-
Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases

1       42420   South  FUR-CH-10000454        Furniture       Chairs

2       90036    West  OFF-LA-10000240  Office Supplies       Labels

3       33311   South  FUR-TA-10000577        Furniture       Tables

4       33311   South  OFF-ST-10000760  Office Supplies      Storage


                                        Product Name     Sales
Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600
2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
```

```
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                    Eldon Fold 'N Roll Cart System   22.3680
2

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164

[5 rows x 21 columns]

df

      Row ID        Order ID  Order Date   Ship Date       Ship
Mode  \
0          1  CA-2016-152156   11/8/2016  11/11/2016    Second Class

1          2  CA-2016-152156   11/8/2016  11/11/2016    Second Class

2          3  CA-2016-138688   6/12/2016   6/16/2016    Second Class

3          4  US-2015-108966  10/11/2015  10/18/2015  Standard Class

4          5  US-2015-108966  10/11/2015  10/18/2015  Standard Class

...      ...             ...         ...         ...             ...

9989    9990  CA-2014-110422   1/21/2014   1/23/2014    Second Class

9990    9991  CA-2017-121258   2/26/2017    3/3/2017  Standard Class

9991    9992  CA-2017-121258   2/26/2017    3/3/2017  Standard Class

9992    9993  CA-2017-121258   2/26/2017    3/3/2017  Standard Class

9993    9994  CA-2017-119914    5/4/2017    5/9/2017    Second Class


     Customer ID     Customer Name    Segment        Country
City  \
0       CG-12520       Claire Gute   Consumer  United States
Henderson
1       CG-12520       Claire Gute   Consumer  United States
Henderson
2       DV-13045   Darrin Van Huff  Corporate  United States      Los
Angeles
3       SO-20335    Sean O'Donnell   Consumer  United States  Fort
```

```
Lauderdale
4       SO-20335    Sean O'Donnell   Consumer  United States  Fort
Lauderdale
...          ...               ...        ...            ...
...
9989    TB-21400  Tom Boeckenhauer   Consumer  United States
Miami
9990    DB-13060       Dave Brooks   Consumer  United States
Costa Mesa
9991    DB-13060       Dave Brooks   Consumer  United States
Costa Mesa
9992    DB-13060       Dave Brooks   Consumer  United States
Costa Mesa
9993    CC-12220      Chris Cortes   Consumer  United States
Westminster

      ... Postal Code  Region       Product ID         Category Sub-
Category  \
0     ...       42420   South  FUR-BO-10001798        Furniture
Bookcases
1     ...       42420   South  FUR-CH-10000454        Furniture
Chairs
2     ...       90036    West  OFF-LA-10000240  Office Supplies
Labels
3     ...       33311   South  FUR-TA-10000577        Furniture
Tables
4     ...       33311   South  OFF-ST-10000760  Office Supplies
Storage
...   ...         ...     ...              ...              ...
...
9989  ...       33180   South  FUR-FU-10001889        Furniture
Furnishings
9990  ...       92627    West  FUR-FU-10000747        Furniture
Furnishings
9991  ...       92627    West  TEC-PH-10003645       Technology
Phones
9992  ...       92627    West  OFF-PA-10004041  Office Supplies
Paper
9993  ...       92683    West  OFF-AP-10002684  Office Supplies
Appliances

                                             Product Name     Sales
Quantity  \
0                     Bush Somerset Collection Bookcase  261.9600
2
1     Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2     Self-Adhesive Address Labels for Typewriters b...   14.6200
2
```

```
3         Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                         Eldon Fold 'N Roll Cart System   22.3680
2
...                                                    ...       ...
...
9989                              Ultra Door Pull Handle   25.2480
3
9990  Tenex B1-RE Series Chair Mats for Low Pile Car...   91.9600
2
9991                               Aastra 57i VoIP phone  258.5760
2
9992  It's Hot Message Books with Stickers, 2 3/4" x 5"   29.6000
4
9993  Acco 7-Outlet Masterpiece Power Center, Wihtou...  243.1600
2

      Discount    Profit
0         0.00   41.9136
1         0.00  219.5820
2         0.00    6.8714
3         0.45 -383.0310
4         0.20    2.5164
...        ...       ...
9989      0.20    4.1028
9990      0.00   15.6332
9991      0.20   19.3932
9992      0.00   13.3200
9993      0.00   72.9480

[9994 rows x 21 columns]
```

Let’s start by looking at the descriptive statistics of the dataset

```
df.describe()

            Row ID   Postal Code         Sales     Quantity
Discount  \
count  9994.000000   9994.000000   9994.000000  9994.000000
9994.000000
mean   4997.500000  55190.379428    229.858001     3.789574
0.156203
std    2885.163629  32063.693350    623.245101     2.225110
0.206452
min       1.000000   1040.000000      0.444000     1.000000
0.000000
```

```
25%    2499.250000  23223.000000     17.280000     2.000000
0.000000
50%    4997.500000  56430.500000     54.490000     3.000000
0.200000
75%    7495.750000  90008.000000    209.940000     5.000000
0.200000
max    9994.000000  99301.000000  22638.480000    14.000000
0.800000

            Profit
count  9994.000000
mean     28.656896
std     234.260108
min   -6599.978000
25%       1.728750
50%       8.666500
75%      29.364000
max    8399.976000

#The dataset has an order date column.  used this column to create new
columns like order month, order year, and order day, which will be
very valuable for sales and profit analysis according to time periods.


df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
```

```
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

Converting Date Columns

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Ship Date'] = pd.to_datetime(df['Ship Date'])
# Date Conversion: The Order Date and Ship Date columns have been
converted into datetime format for date-based analysis.”

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   datetime64[ns]
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: datetime64[ns](2), float64(3), int64(3), object(13)
memory usage: 1.6+ MB

df.head()

   Row ID        Order ID Order Date  Ship Date       Ship Mode
Customer ID  \
0       1  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
```

```
12520
1       2  CA-2016-152156 2016-11-08 2016-11-11    Second Class    CG-
12520
2       3  CA-2016-138688 2016-06-12 2016-06-16    Second Class    DV-
13045
3       4  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335
4       5  US-2015-108966 2015-10-11 2015-10-18  Standard Class    SO-
20335

     Customer Name    Segment        Country             City  ...  \
0      Claire Gute   Consumer  United States        Henderson  ...
1      Claire Gute   Consumer  United States        Henderson  ...
2  Darrin Van Huff  Corporate  United States      Los Angeles  ...
3   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...
4   Sean O'Donnell   Consumer  United States  Fort Lauderdale  ...

  Postal Code  Region       Product ID         Category Sub-
Category  \
0       42420   South  FUR-BO-10001798        Furniture    Bookcases

1       42420   South  FUR-CH-10000454        Furniture       Chairs

2       90036    West  OFF-LA-10000240  Office Supplies       Labels

3       33311   South  FUR-TA-10000577        Furniture       Tables

4       33311   South  OFF-ST-10000760  Office Supplies      Storage


                                        Product Name     Sales
Quantity  \
0                  Bush Somerset Collection Bookcase  261.9600
2
1  Hon Deluxe Fabric Upholstered Stacking Chairs,...  731.9400
3
2  Self-Adhesive Address Labels for Typewriters b...   14.6200
2
3      Bretford CR4500 Series Slim Rectangular Table  957.5775
5
4                     Eldon Fold 'N Roll Cart System   22.3680
2

   Discount    Profit
0      0.00   41.9136
1      0.00  219.5820
2      0.00    6.8714
3      0.45 -383.0310
4      0.20    2.5164
```

```
[5 rows x 21 columns]
```

Adding New Date-Based Columns

```
df['Order Month'] = df['Order Date'].dt.month
df['Order Year'] = df['Order Date'].dt.year
df['Order Day of Week'] = df['Order Date'].dt.dayofweek
```

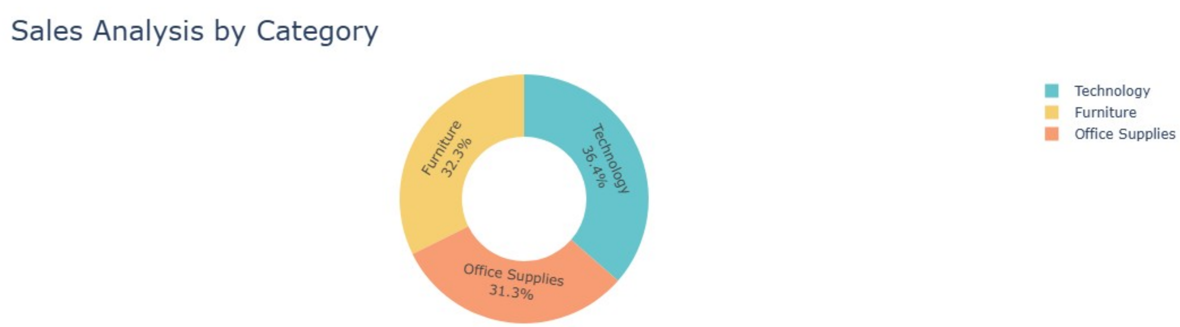
Monthly Sales Analysis

```
sales_by_month = df.groupby('Order Month')
['Sales'].sum().reset_index()
fig = px.line(sales_by_month,
              x='Order Month',
              y='Sales',
              title='Monthly Sales Analysis')
fig.show()
```


```
#Data Grouping:

#data.groupby('Order Month')['Sales'].sum() is used to calculate the
total sales for each month.

#reset_index() organizes the grouped data into a clean, structured
format.

#px.line: A line chart is created to show the monthly sales trend.

#fig.show(): Displays the graph.
```

Sales Analysis by Category

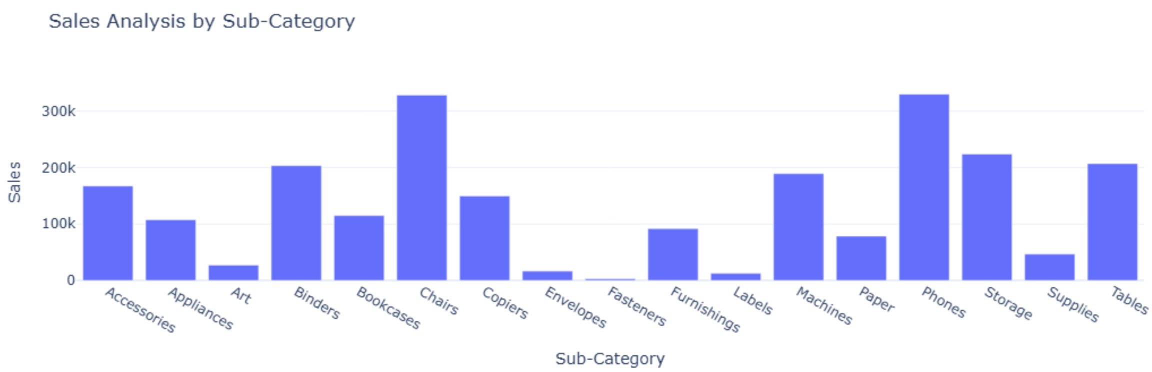
```
sales_by_category = df.groupby('Category')
['Sales'].sum().reset_index()


fig = px.pie(sales_by_category,
             values='Sales',
             names='Category',
             hole=0.5,
             color_discrete_sequence=px.colors.qualitative.Pastel)

fig.update_traces(textposition='inside', textinfo='percent+label')
fig.update_layout(title_text='Sales Analysis by Category',
title_font=dict(size=24))

fig.show()
```




```
#groupby('Category'): Extracts category-wise sales.


#Pie Chart:


#px.pie: Displays sales proportions in a pie chart.


#hole=0.5: Creates a donut-style chart.


#Pastel Colors: A soft color palette is used in the chart.
```

Sales Analysis by Sub-category

```
sales_by_subcategory = df.groupby('Sub-Category')
['Sales'].sum().reset_index()
fig = px.bar(sales_by_subcategory,
             x='Sub-Category',
             y='Sales',
             title='Sales Analysis by Sub-Category')
fig.show()
```



Monthly Profit Analysis

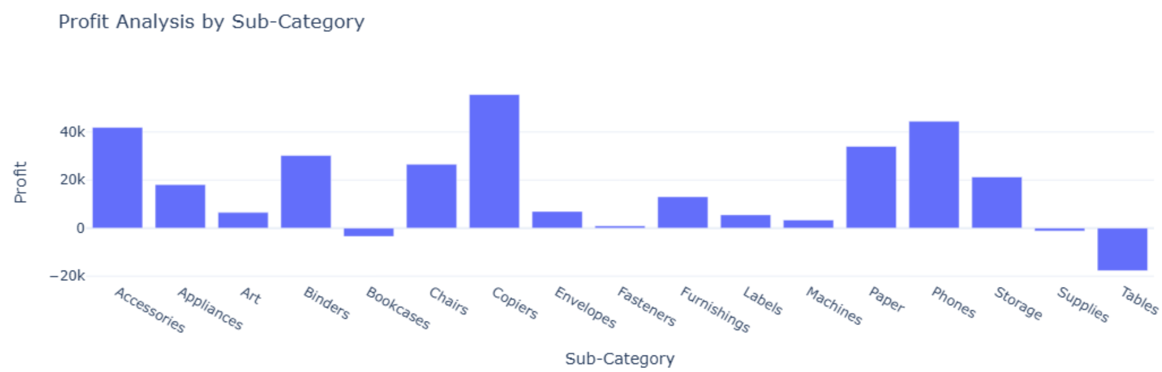
```
profit_by_month = df.groupby('Order Month')
['Profit'].sum().reset_index()

profit_by_month = df.groupby('Order Month')
['Profit'].sum().reset_index()
fig = px.line(profit_by_month,
              x='Order Month',
              y='Profit',
              title='Monthly Profit Analysis')
fig.show()
```

Profit Analysis by Category

```
profit_by_category = df.groupby('Category')
['Profit'].sum().reset_index()

fig = px.pie(profit_by_category,
             values='Profit',
             names='Category',
             hole=0.4,
             color_discrete_sequence=px.colors.qualitative.Pastel)

fig.update_traces(textposition='inside', textinfo='percent+label')
fig.update_layout(title_text='Profit Analysis by Category',
title_font=dict(size=24))

fig.show()
```

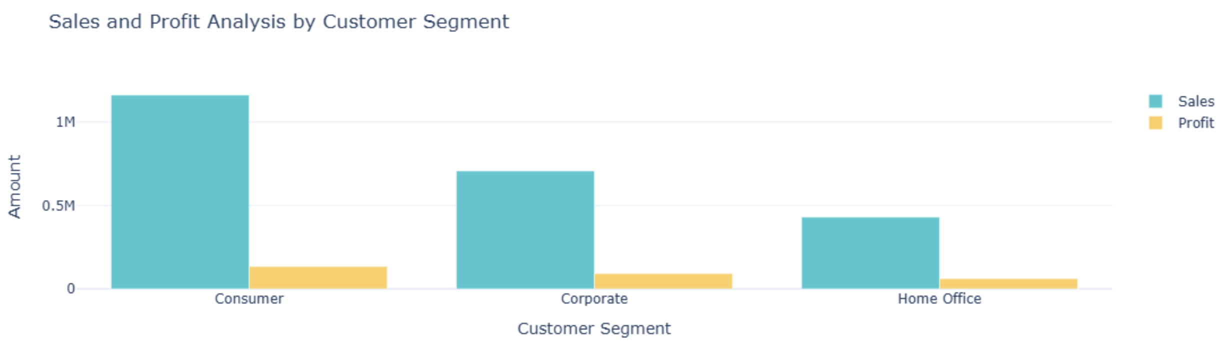
Profit Analysis by Sub-Category

```
profit_by_subcategory = df.groupby('Sub-Category')
['Profit'].sum().reset_index()
fig = px.bar(profit_by_subcategory, x='Sub-Category',
             y='Profit',
             title='Profit Analysis by Sub-Category')
fig.show()
```



Sales and Profit Analysis by Customer Segment

```
sales_profit_by_segment = df.groupby('Segment').agg({'Sales': 'sum',
'Profit': 'sum'}).reset_index()

color_palette = colors.qualitative.Pastel

fig = go.Figure()
fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Sales'],
                     name='Sales',
                     marker_color=color_palette[0]))

fig.add_trace(go.Bar(x=sales_profit_by_segment['Segment'],
                     y=sales_profit_by_segment['Profit'],
                     name='Profit',
                     marker_color=color_palette[1]))

fig.update_layout(title='Sales and Profit Analysis by Customer
Segment',
                  xaxis_title='Customer Segment',
yaxis_title='Amount')

fig.show()
```



analyse sales-to-profit ratio

```
sales_profit_by_segment = df.groupby('Segment').agg({'Sales': 'sum',
'Profit': 'sum'}).reset_index()
sales_profit_by_segment['Sales_to_Profit_Ratio'] =
sales_profit_by_segment['Sales'] / sales_profit_by_segment['Profit']
print(sales_profit_by_segment[['Segment', 'Sales_to_Profit_Ratio']])
```

```
       Segment  Sales_to_Profit_Ratio
0     Consumer               8.659471
1    Corporate               7.677245
2  Home Office               7.125416
```

Conclusion

```
#Optimizing high-margin products, managing low-performing categories,
and aligning strategies with seasonal trends can improve profitability
and stabilize sales.
```